- Modulation d'amplitude | - PC - SM | 2 BAC - Biof -

Signal modulant :

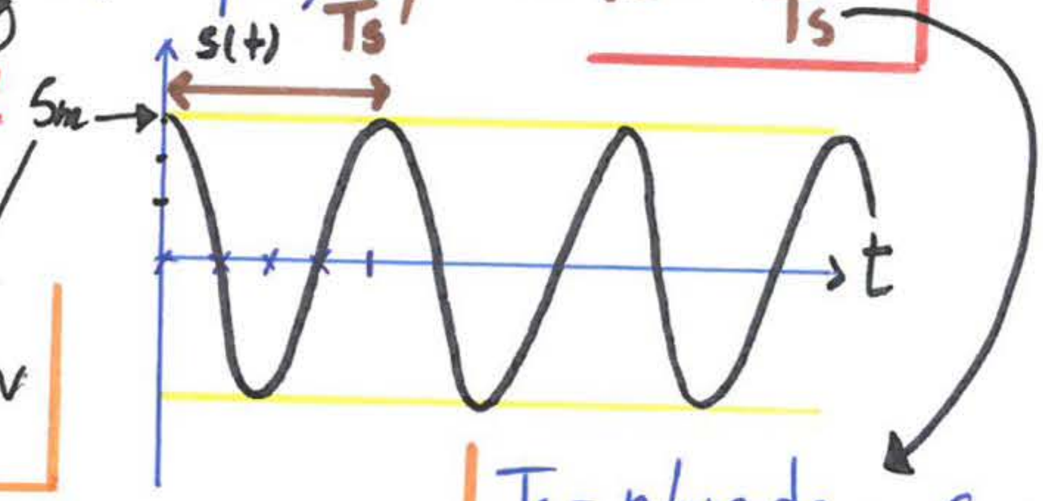
- السلام عليكم → Son → Mic → Tension électrique

$$s(t) = S_m \cdot \cos(2\pi f_s t)$$

$$f_s = \frac{1}{T_s}$$

S_m = nbre de div × S_V

T_s = nbre de div × S_H

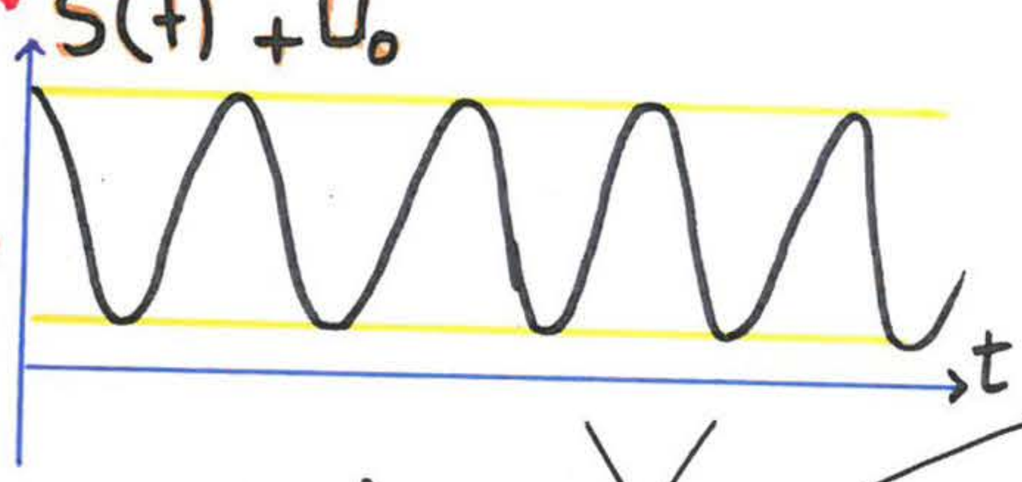


⚠ f_s : faible fréquence

+

U_0 : tension Continue de décalage

$$s(t) + U_0$$

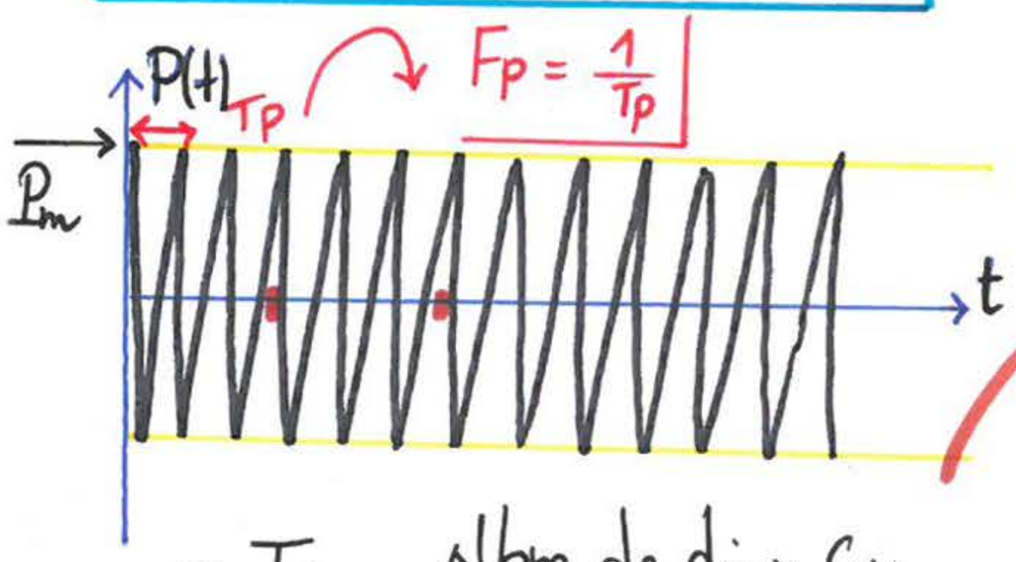


Onde porteuse P(t)

⚠ Fp : Haute fréquence

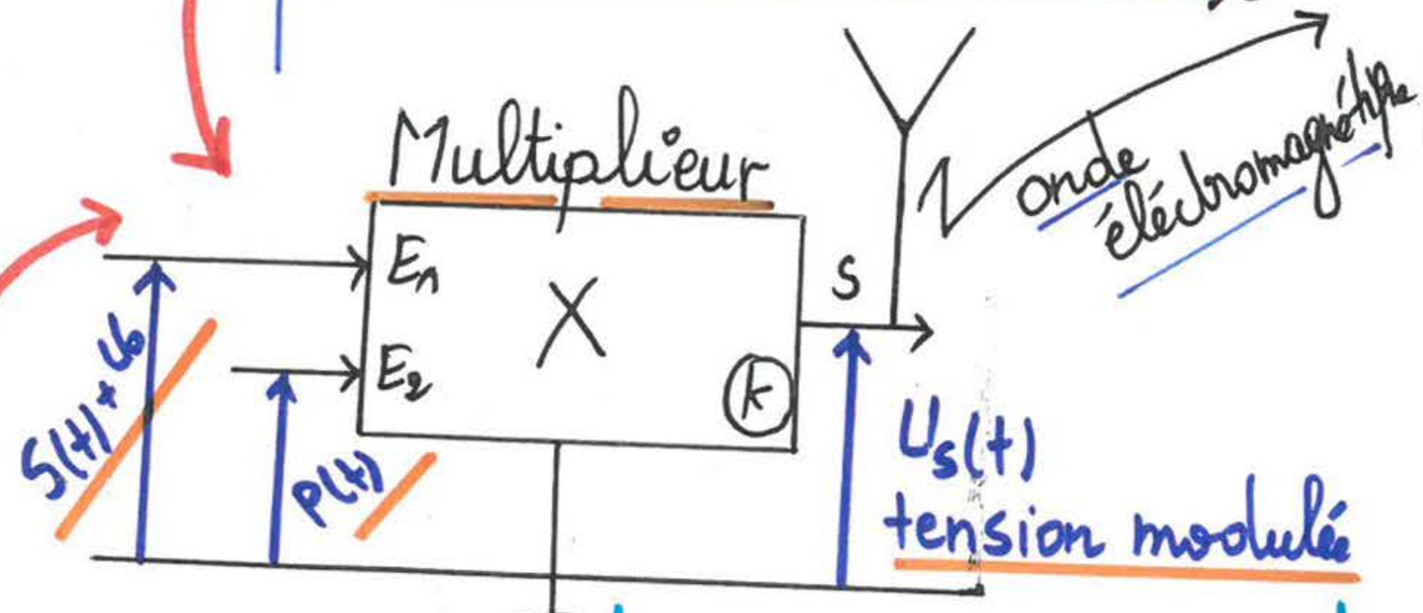
$$P(t) = P_m \cdot \cos(2\pi F_p \cdot t)$$

$$F_p = \frac{1}{T_p}$$

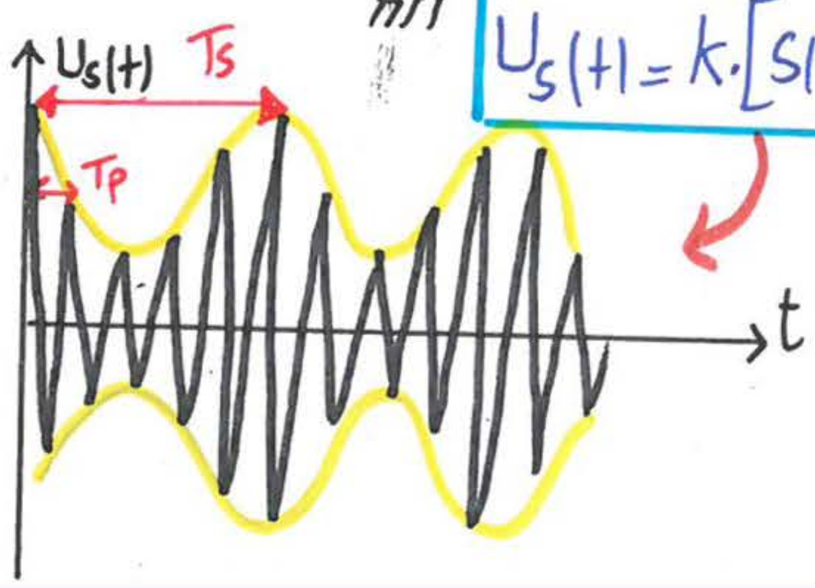


$x \cdot T_p$ = Nbre de div × S_H

$$T_p = \frac{\text{Nbre de div} \times S_H}{x}$$



$$U_s(t) = k \cdot [s(t) + U_0] \times P(t)$$



Q1 Determiner l'unité de K (cste de multiplication)

On sait que :

$$U'_s(t) = K\,\underbrace{[S(t)+U_0]}_{U_1}\cdot\underbrace{P(t)}_{U_2}$$

$$U'_s(t) = K \cdot U_1 \cdot U_2$$

$$\boxed{[K] = \frac{[U]}{[U]^2} = [U]^{-1}}$$

Donc : K en V^{-1}.

Q2 Montrer que : $U_s(t) = A\left(m.\cos(2\pi f_s t)+1\right).\cos(2\pi F_p t)$

On sait que :

Données ✓

$$U_s(t) = K.\left(S(t)+U_0\right).P(t)$$

$$U_s(t) = K.\left(S_m.\cos(2\pi f_s t)+U_0\right).P_m.\cos(2\pi F_p t)$$

$$U_s(t) = \underbrace{K.P_m.U_0}_{A}\left(\underbrace{\frac{S_m}{U_0}}_{m}.\cos(2\pi f_s t)+1\right).\cos(2\pi F_p t)$$

$$\boxed{U_s(t) = A.\left(m.\cos(2\pi f_s t)+1\right).\cos(2\pi F_p t)}$$

Tension modulée

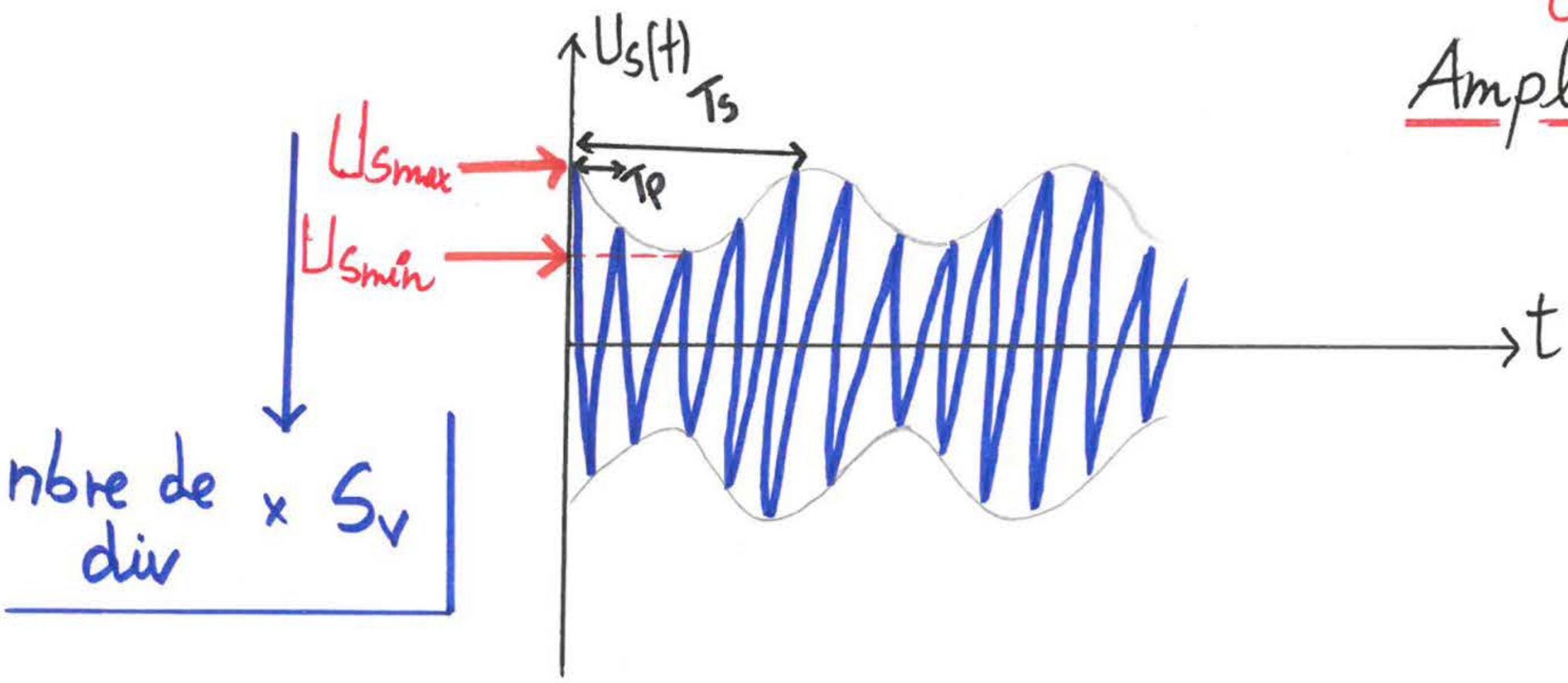
avec : $\begin{cases} A = K.P_m.U_0 \\ m = \dfrac{S_m}{U_0} \end{cases}$

taux de modulation

⚠ Tension modulée : $U_s(t) = A\left(m \cdot \cos(2\pi f_s t) + 1\right) \cdot \cos(2\pi f_p t)$

⚠ Amplitude de la tension modulée : $U_{sm} = A\left(m \cdot \cos(2\pi f_s t) + 1\right)$

On dit : Amplitude Variable



Q3 | Exprimer U_{smax}, U_{smin} :

On a : $U_{sm} = A\left(m \cdot \cos(2\pi f_s t) + 1\right)$

par encadrement :

$$-1 \leqslant \cos(2\pi f_s t) \leqslant 1$$

$$1 - m \leqslant 1 + m \cdot \cos(2\pi f_s t) \leqslant 1 + m$$

$$\underbrace{A(1-m)}_{U_{smin}} \leqslant U_{sm} \leqslant \underbrace{A(1+m)}_{U_{smax}}$$

Q4 | Taux de modulation m :

$$\boxed{\cdot\ m = \frac{S_m}{U_0}} \quad (1)$$

\# $U_{Smax} - U_{Smin} = 2.A.m$

$U_{Smax} + U_{Smin} = 2.A$

$$\longrightarrow \boxed{m = \frac{U_{Smax} - U_{Smin}}{U_{Smax} + U_{Smin}}} \quad (2)$$

Q5 | Conditions de Bonne modulation :

$$\boxed{\begin{array}{l} \cdot\ m < 1 \\ \cdot\ F_p \geqslant 10.f_s \end{array}}$$

Q6 | Définition : Modulation d'amplitude :

C'est rendre l'amplitude de signal modulé une fonction Affine de la tension modulante.